

Electrolysis: Splitting Water with Electricity

1. What if Electricity Could BREAK Water?

Imagine filling up a clear glass with pure, clean water. It looks completely still and simple, right? But inside that glass lies a microscopic world full of magic! What if you could use the invisible power of electricity to actually reach inside and rip water apart? When we run a small, safe electrical current through water, something amazing happens: tiny bubbles start dancing and rising to the surface. These bubbles are signs that water is no longer just water anymore—it is breaking down into two completely different gases!

2. Water's Secret Recipe

Just like a delicious batch of roti dough is mixed using specific amounts of atta and water, nature has a very strict secret recipe for making water. Water is built out of incredibly tiny blocks called **molecules**. Every single molecule of water is made up of exactly two parts of **Hydrogen** bonded tightly to one part of **Oxygen**. Because of this special atomic combination, scientists around the world use the famous shorthand chemical formula: H_2O .



Scientific Vocabulary Check:

Molecule: The smallest possible particle of a substance that still keeps all the properties of that substance. If you split a water molecule any further, it stops being water!

3. Breaking the Water Molecule

The word **Electrolysis** might sound long and complicated, but it is actually a combination of two simple Greek words: *Electro* (meaning electricity) and *Lysis* (meaning to cut apart or untie). So, electrolysis literally means using the power of an electric current like a pair of invisible microscopic scissors to cut the strong chemical bonds holding the H_2O molecule together!

When these bonds are cut, the atoms separate and transform into two distinct gases: pure Hydrogen gas (H_2) and pure Oxygen gas (O_2).

4. The Crucial Helpers: Electrodes and Electrolytes

To successfully run this experiment at home or in the classroom, electricity needs a proper doorway to enter and leave the water. This is where our tools come in:

- **Electrodes (Pencils Are Important!):** An electrode is simply a solid conductor that lets electricity travel into or out of a liquid. In our experiment, we use two ordinary graphite pencils sharpened at both ends. Think of them like the handles of a hot kitchen pan—one pencil connects to the positive (+) side of our power source, and the other connects to the negative (-) side.
- **Electrolyte (Salt Helps!):** Pure water by itself is actually quite stubborn and does not let electricity pass through it easily. To help the electricity flow smoothly, we mix in a spoonful of ordinary table salt. The salt acts as a magical bridge booster. Any substance dissolved in water to help it conduct electricity is called an **Electrolyte**.

5. What Happens at the Electrodes?

If you look very closely at the submerged pencil tips while the electricity is running, you will notice a fascinating difference between them:

- At the **Negative Electrode** (called the Cathode), **Hydrogen gas** forms. Because every water molecule contains twice as much Hydrogen as Oxygen, you will see a massive rush of active, bubbling gas forming here!
- At the **Positive Electrode** (called the Anode), **Oxygen gas** forms. Since there is less oxygen in the secret recipe, this pencil tip will have significantly fewer bubbles rising from it.

Safety First!

Always make sure an adult or teacher is supervising your science experiments. **Only use a standard 9-volt (9V) battery** for this activity. Never, ever touch exposed wires, and never try to plug any experimental wires into the electrical wall sockets at home or school!

6. Step-by-Step Experiment Guide

01

Fill a clean, clear glass about three-quarters full with ordinary room-temperature water.

02

Add one spoonful of common table salt into the glass and stir thoroughly until it fully disappears.

03

Carefully connect two electrical connection wires to the positive and negative terminals of a 9V battery.

04

Attach the other ends of the wires to the exposed graphite tips of two pencils, and dip the opposite sharp ends into the saltwater solution. Watch the bubbles appear!

7. Real-World Wonders of Electrolysis in India

Electrolysis isn't just a cool classroom trick; it is changing how people live across India right now! Modern eco-friendly green hydrogen buses are currently being tested on the busy streets of Delhi and Mumbai to reduce pollution. Huge industrial electrolysis plants manufacture vital oxygen cylinders used daily in local hospitals. The dazzling gold-plated jewellery worn at beautiful Indian weddings gets its shiny coating using a similar electrical process. Even the high-tech Indian Navy submarine, *INS Arihant*, relies on automated electrolysis systems deep underwater to create fresh, breathable oxygen for its brave sailors!

Student Practice Worksheet

Answer the following questions carefully based on what you learned in this lesson. Write your answers neatly in your notebook.

1. **The word "Electrolysis" can be split into two smaller words that explain what it does. What do those two parts mean in simple words?**
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2. **Write down the famous three-character chemical formula for a single water molecule, and explain what each letter stands for.**
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3. **According to nature's "secret recipe" for water, how many Hydrogen atoms combine with exactly one Oxygen atom?**
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4. **Why does pure water need us to add a spoonful of common table salt before the electrolysis experiment can begin?**
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5. **What is the special scientific term used to describe a liquid or substance (like salt water) that helps conduct electricity?**
.....
6. **In our classroom lab setup, what everyday school supply item did we use to act as the two "electrodes"?**
.....
7. **Imagine you are looking closely at the two pencils in the glass. One pencil tip is covered in way more bubbles than the other. Which gas is forming these extra bubbles?**
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8. At which electrode (the Positive electrode or the Negative electrode) does the precious Oxygen gas collect?

9. State the critical safety rule regarding the type of battery you are allowed to use for this science experiment.

10. Name two major Indian cities where green hydrogen made via electrolysis is helping power public eco-buses.

11. How does the Indian Navy submarine *INS Arihant* use the science of electrolysis to protect its brave crew members deep under the sea?

12. Explain in your own words what happens to a water molecule when the invisible "electric scissors" cut it apart.
